Time Value of Money (TVM)
Detailed Lecture Notes with Solved Examples

I. Fundamental Principle: Time Value of Money

Core Idea

Money has a **time-dependent value**.

$$1 \text{ dollar today } > 1 \text{ dollar in the future}$$

because today's money can be **invested and earn return**.

Economic Logic

- Investment generates interest or return.
- Inflation reduces purchasing power.
- Future payments involve risk and uncertainty.

Financial proverb: "A bird in hand is better than two in the bush."

Key Definitions

Term	**Meaning**
Present Value (PV)	Value of money at time $t = 0$
Future Value (FV)	Value of money at a future time
Compounding	Moving PV to FV
Discounting	Moving FV to PV

II. Mathematical Formulas for Single Amounts

Future Value

$$FV = PV(1 + i)^n$$

Present Value

$$PV = \frac{FV}{(1+i)^n}$$

Example: Simple Investment

Given:

- $PV = 100$
- $i = 5\%$
- $n = 1$ year

$$FV = 100(1.05) = 105$$

After two years:

$$FV = 100(1.05)^2 = 110.25$$

Insight: Interest in the second year is earned on \$105, not \$100 $\Rightarrow$ compound interest.

III. Compounding Frequency

Interest may compound annually, semi-annually, quarterly, or monthly.

General Formula

$$FV = PV\left(1 + \frac{i}{m}\right)^{nm}$$

where m is the number of compounding periods per year.

Example: Monthly Compounding

Given:

- $PV = 2235$
- $i = 2\%$
- $m = 12$
- $n = 4$ years

$$FV = 2235\left(1 + \frac{0.02}{12}\right)^{48}$$

$$FV \approx 2420.98$$

IV. Financial Markets Context

Money Market: Trades short-term securities (maturity < 1 year), such as:

- Treasury Bills
- Commercial Paper
- Certificates of Deposit

V. Decision Case Study: Car Financing

Car price = \$20,000, market interest rate = 10%.

Option A (Credit)

- Pay \$8,000 today
- Pay \$12,000 after 2 years

Present value of future payment:

$$PV = \frac{12{,}000}{(1.10)^2} = 9{,}917$$

Total present value:

$$PV_A = 8{,}000 + 9{,}917 = 17{,}917$$

Option B (Cash Discount)

$$PV_B = 19{,}000$$

Decision

$$17{,}917 < 19{,}000 \Rightarrow \text{Option A is better.}$$

VI. Real-World Insight: "Interest-Free" Offers

Companies recover hidden costs through:

- Higher base prices
- Opportunity cost of capital
- Compensating bank balances

Conclusion: Nothing is truly free in finance.

VII. Annuities

Definition

An annuity is a stream of **equal periodic payments**.

Types

Type	Payment Timing
Ordinary Annuity	End of each period
Annuity Due	Beginning of each period

Relationship

$$FV_{\text{Annuity Due}} = FV_{\text{Ordinary Annuity}}(1+i)$$

VIII. Financial Calculator Variables

Button	Meaning
N	Number of periods
I/Y	Interest rate
PV	Present value
PMT	Payment
FV	Future value

IX. Case Study: Lottery Choice

Options:

- Lump sum: \$1.3 million today
- Annuity: \$100,000 per year for 25 years at 5%

Present Value of Annuity

$$PV = PMT \times \frac{1-(1+i)^{-n}}{i}$$

$$PV = 100{,}000 \times \frac{1-(1.05)^{-25}}{0.05}$$

$$PV \approx 1{,}409{,}000$$

Decision

$$1.409\text{M} > 1.3\text{M} \Rightarrow \text{Annuity is better.}$$

X. Key Conceptual Summary

- Compare all cash flows at the **same point in time**.
- Compounding increases future value.
- Discounting finds true present worth.
- Earlier cash flows are more valuable.
- Annuity Due has higher value than Ordinary Annuity.
- Real-world finance includes hidden interest costs.

XI. Exam Takeaways

Must-Know Formulas

$$FV = PV(1+i)^n$$
$$PV = \frac{FV}{(1+i)^n}$$
$$FV = PV\left(1+\frac{i}{m}\right)^{nm}$$
$$PV_{\text{annuity}} = PMT\frac{1-(1+i)^{-n}}{i}$$
$$FV_{\text{AD}} = FV_{\text{OA}}(1+i)$$

Problem-Solving Strategy

1. Draw a timeline.
2. Identify PV, FV, i, n, and PMT.
3. Convert all cash flows to the same date.
4. Choose the highest value or lowest cost.